Joseph Robert Livesey

Department of Astronomy, University of Wisconsin–Madison

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Research Interests

Dynamics of planetary systems, star clusters, and AGN; exoplanet demography and orbital architectures of multi-planet systems; planetary formation, oceans, interiors, atmospheres, and habitability.

Education

PhD **University of Wisconsin–Madison**, Astronomy — Expected 2028

- Advisor: Juliette Becker
- Doctoral minor: Atmospheric & Oceanic Sciences

MS **University of Wisconsin–Madison**, Astronomy — 2025

BS **University of Washington**, Astronomy and Physics — 2022

- Departmental honors in Astronomy and in Physics
- Minors: Linguistics and Mathematics

Research Appointments

Graduate Research Assistant, University of Wisconsin–Madison	2023–2028
Research Analyst, University of Washington	2023

Honors and Awards

Graduate & Professional Research Fellowship, Wisconsin Space Grant ($6,000)	2025
Blair Savage Travel Award, U. Wisconsin	2025
Fluno Graduate Fellowship, U. Wisconsin	2023
President's Fellowship, Georgia Institute of Technology (*declined*)	2023
Phi Beta Kappa	2022
Mary Gates Research Scholarship, U. Washington ($5,000)	2020

Teaching Experience

U. Wash.	TA, ASTR 150 (The Planets)	Spring 2023
	Grader, ASTR 427 (Computational Methods)	Autumn 2022
	Grader, ASTR 150 (The Planets)	Winter 2023
	Grader, ASTR/ASTBIO 115 (Life in the Universe)	Spring 2022
	Grader, ASTR 101	Winter 2022, Autumn 2022

Service to the Profession

Executive Secretary, NASA XRP	1 panel
AbSciCon Primary Convener, Session: "Dynamical Environments of Habitable Worlds"	May 2026
Local Organizing Committee, Great Lakes Exoplanet Area Meeting (GLEAM)	Nov 2025
Scientific Organizing Committee, VPLANET workshop	Sep 2022

Outreach

Lead Coordinator, Madison Astronomy on Tap	2025–
Coordinator, Madison Astronomy on Tap	2023–2025
Volunteer, Washburn Observatory Public Observing Nights	2023–

Refereed Publications

First-authored: 4. Total: 10. Total citations: 57. $h = 4$. *Citation counts are listed on the left.*

First author

1 "Discovery and Characterization of the TOI-4468 Planetary System: A Transiting Hot Jupiter With a Lone Nearby Outer Companion."
J. R. Livesey, B. J. Hord, J. Becker, A. Vanderburg, J. E. Rodriguez, E. Koo et al., *AJ*, 172, 287.

4 "Tides Tighten the Hycean Habitable Zone."
J. R. Livesey, J. Becker & S. L. Widicus Weaver 2025, *ApJL*, 987, L8.

10 "Secular Perturbations From Exterior Giants Strongly Influence Gap Complexity in Peas-in-a-Pod Exoplanetary Systems."
J. R. Livesey & J. Becker 2025, *ApJ*, 979, 202.

12 "Orbital Stability and Secular Dynamics of the Proxima Centauri Planetary System."
J. R. Livesey, R. Barnes & R. Deitrick 2024, *ApJ*, 964, 4.

Second or third author

– "On the Fate of Exomoons Around Migrating White Dwarf Planets."
R. S. Narayan, **J. R. Livesey** & J. Becker *submitted to ApJL.*

9 "The Fate of Oceans on First-Generation Planets Orbiting White Dwarfs."
J. Becker, A. Vanderburg & **J. R. Livesey** 2025, *ApJ*, 979, 99.

Contributing author

4 "GJ 523 b is a Massive, 220 Myr-old Mega-Earth, Likely on a Polar Orbit."
M. A. Kroft et al. **including J. R. Livesey**, *submitted to AJ.*

4 "THYME. XIII. Two Young Neptunes Orbiting a 75-Myr Star in the Alpha Persei Cluster."
A. Dattilo, A. Vanderburg, M. G. Barber, A. W. Mann, R. Kerr, A. L. Kraus, **J. R. Livesey** et al. 2025, *AJ*, 170, 318.

3 "A Pair of Dynamically Interacting Sub-Neptunes Around TOI-6054."
M. A. Kroft, T. G. Beatty, I. J. M. Crossfield, **J. R. Livesey** et al. 2025, *AJ*, 170, 150.

5 "The Effects of Kinematic MHD on the Atmospheric Circulation of Eccentric Hot Jupiters."
H. Beltz, W. Houck, L. C. Mayorga, T. D. Komacek, **J. R. Livesey** & J. Becker 2025, *ApJ*, 984, 90

5 "History and Habitability of the LP 890-9 Planetary System."
R. Barnes, L. N. R. do Amaral, J. Birky, L. Carone, P. E. Driscoll, **J. R. Livesey** et al. 2025, *PSJ*, 6, 25.

Presentations

Invited talks

4. "The Structures of Compact Multiple-Planet Systems."
Exoplanets Seminar, University of Warwick. 20 October 2026.

3. "The Structures of Compact Multiple-Planet Systems."
Center for Exoplanets & Habitable Worlds Seminar, Pennsylvania State University. 21 September 2026.

2. "Close Outer Companions to Hot Jupiters."
D. Z. Seligman & A. D. Feinstein Group Meeting, Michigan State University. 10 April 2026.

1. "Close Outer Companions to Hot Jupiters."
Star and Planet Formation Journal Club, University of Michigan. 7 April 2026.

Contributed talks

6. "The Resonant Evolution of Hot Jupiters With Their Companion Planets."
Wisconsin Space Grant Conference, Kenosha, WI. August 2026.

5. "Discovery and Dynamics of a Transiting Hot Jupiter and Its Lone Outer Companion."
Great Lakes Exoplanet Area Meeting, Madison, WI. November 2025.

4. "Outer Giants Strongly Influence Gap Complexity in Peas-in-a-Pod Systems."
DDA 56, Atlanta, GA. May 2025.
3. "Quantifying the Surface Heating of Tidally Migrating White Dwarf Planets."
AAS 244, Madison, WI. June 2024.
2. "Orbital Dynamics of the Proxima Centauri System."
AAS 241, Seattle, WA. January 2023.
1. "Orbital and Rotational Evolution of Proxima Centauri b."
AAS 237, virtual. January 2021.

Posters

4. "The Dearth of Nearby Outer Companions to Hot Jupiters."
DDA 57, Chicago, IL. June 2026.
3. "Tides Tighten the Hycean Habitable Zone."
Astrobiology Science Conference 2026, Madison, WI. May 2026.
2. "Outer Giants Strongly Influence the Gap Complexity of Tightly Packed Exoplanetary Systems."
WiCOR Kickoff Poster Session, Madison, WI. September 2024.
1. "Quantifying the Surface Heating of Tidally Migrating White Dwarf Planets."
Astrobiology Science Conference 2024, Providence, RI. May 2024.

Public talks

1. "The Ninth Planet(s) of the Solar System, and Why Pluto Had to Die."
Astronomy on Tap, Madison, WI. 17 Feb 2026.

Telescope Time Awarded

Confirming TOI-4468.03: An Outer Giant in an Atypical Hot Jupiter System 2026B
PI. WIYN 3.5-meter – 8 hours with NEID spectrograph

Unveiling the Nature of a Potential Hycean World: TOI-7032.01 2025A
Co-I. WIYN 3.5-meter – 10 hours with NEID spectrograph

Hot Jupiter Formation Pathways: Mass Measurements for a New Multi-Planet System 2024A, 2025A
Co-I. WIYN 3.5-meter – 13 hours with NEID spectrograph

Professional Development

Planets in the Galactic Context Workshop, New York, NY Sept 2026
Exoplanets: Compositions, Mineralogy, Evolution (RiMG workshop), Chicago, IL Aug 2024

Memberships

- American Astronomical Society (AAS)
- AAS Division on Dynamical Astronomy (DDA)
- AAS Division for Planetary Sciences (DPS)

Technical Skills

Code: Python, C, Julia
Software: Unix, Git, LaTeX, Jupyter, Mathematica, Condor, Slurm, MySQL, HTML, CSS

Graduate Coursework

Astronomy: Observational Techniques, Radiative Processes, Astrophysical Fluids, Stellar Astrophysics, Interstellar Medium, Galaxies, Cosmology, Solar System Dynamics
Other: Atmospheric Thermodynamics, Physical Oceanography, Large-Scale Ocean–Atmosphere Coupling, Prebiotic Chemistry & Astrobiology

Non-Academic Employment

Library Technician, Odegaard Undergraduate Library	June–Sept 2022
Access Services Student Assistant, Odegaard Undergraduate Library	2018–2022
Front End Assistant, Costco Wholesale	July–Sept 2018

References

Prof. **Juliette Becker** • U. Wisconsin • Doctoral advisor	juliette.becker@wisc.edu
Prof. **Rory Barnes** • U. Washington • Undergraduate research supervisor	rkb9@uw.edu
Prof. **Nicholas C. Stone** • U. Wisconsin	nstone2@wisc.edu